Corrected MCQs

1. **B.** $P_A(B)$ is the probability of B knowing A, for $P(A) > 0$. Intersection and conditioning Inverse are different.
2. **B.** $P(A \cap B) = P(A)P_A(B) = 0{,}4 \times 0{,}5 = 0{,}2$. Successive probabilities are multiplied, they are not multiplied Don't add up.
3. **C.** By definition of independence, $P(A \cap B) = P(A)P(B)$. An empty intersection corresponds to Incompatibility, not independence in general.
4. **B.** "No more than 3" includes 3: $X \leq 3$. "less than 3" would exclude.
5. **B.** The probability of the path is the product of successive conditional probabilities. then disjointed paths to get a meeting.
6. **A.** The number of events is 10, the probability of success 0.3, with independence. counts successes, not probabilities.
7. **B.** A sum of n indicators of success of expectation p has expectation np. The average number failures would be $n(1-p)$.
8. **C.** X is integer: $X > 4$ is equivalent to $X \geq 5$. $X > 5$ wrongly excludes value 5.
9. **C.** Independence is one of Bernoulli's assumptions. His loss no longer allows to automatically apply the model, even if a coincidence of laws may exist in Special cases.
10. **A.** Controlling the candidate rank and the previous one allows to establish the minimality when the probability is monotonous in size. Only one calculation at the candidate rank proves that it is appropriate.

True/false. False: with $P(A) = 1/2$, $P(B) = 1/4$ and $B \subset A$, we have $P_A(B) = 1/2$ and $P_B(A) = 1$. True for X total and k Integer: the two events are complementary. False: a single throw of a coin Balanced has a frequency 0 or 1, not 1/2.

Corrected applications and support

Application 1. Line A: 12 and 588; line B: 20 and 380. $P(D) = 32/1000 = 0{,}032 = 0{,}6 \times 0{,}02 + 0{,}4 \times 0{,}05$. $P_A(D) = 0{,}02 \neq 0.032$: no independence.

Application 2. $0{,}14 = 0{,}4 \times 0{,}2 + 0{,}6x$ gives $x = 0{,}1$. $P_D(A) = 0{,}08/0{,}14 = 4/7$.

Implementation 3. Diseases: 90 positives and 10 negatives; non-malads: 90 positives and 810 negatives. $P_M(+) = 0.9$, but $P_+(M) = 90/180 = 0{,}5$.

Implementation 4. The first one agrees if we have a fixed outcome. Without remission, the probabilities
The number of essays to the first success is not independent.
not fixed: this is not a number of successes in a fixed length scheme.

Application 5. $1 - P(X \leq 2)$; $P(X \leq 2)$; $P(X \leq 5) - P(X \leq 1)$. $P(X = 0) = (7/10)^8 = 0{,}05764801$.

Implementation 6. $q(p) = 3p^2(1-p) + p^3 = 3p^2 - 2p^3$. $q' = 6p(1-p) > 0$ on $]0;1[$, therefore q is strictly increasing on $[0;1]$. Continuity and $q(0) = 0 < 0{,}9 < q(1) = 1$ give existence and uniqueness. $q(0{,}80) = 0{,}896 < 0{,}9 < 0{,}905418 = q(0{,}81)$.

Institutionalization

Keep the table next to the tree; ask at each step "from who?".
integers on an axis before calculating. In parametric problem,
distinguishing the variable X, integer $n = 3$ and parameter p.

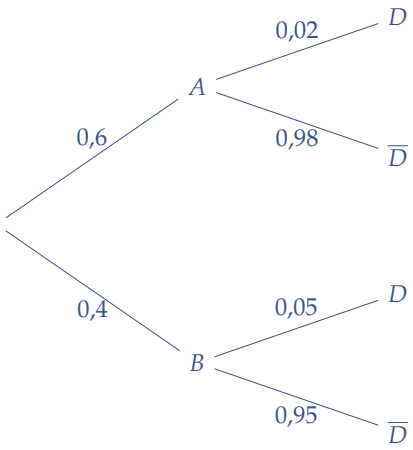
Conduct and correction

Benchmarks for the teacher

Reactivating conditionals in S36 before any Bernoulli schema.
model hypothesis, it does not arise only from repetition. For a draw without discount,
do not automatically invoke a binomial distribution. The formulas of expectation and variance
are used in S39; their evidence in sum of indicators is given in Chapter 14.
requested in December must be able to be obtained by a loop, the logarithm being not yet
taught.

Corrected activity

1. The tree carries the branches A and B, probability 0 ,6 and 0,4. Since A, branches $D, \overline{D}$ carry 0.02, 0.98; since B, they carry 0.05, 0.95.



Total probabilities, $P(D) = P(A)P_A(D) + P(B)P_B(D) = 0{,}012 + 0{,}020 = 0{,}032$.

2. $P(D) > 0$ and $P(A \cap D) = 0{,}6 \times 0{,}02 = 0.012$. Therefore $P_D(A) = \frac{P(A\cap D)}{P(D)} = \frac{0{,}012}{0{,}032} = \frac{3}{8} = 0{,}375$.

3. Each sample is a test with two exits, "failure" or "non-failure", with the
the same probability 0.032 of defect. The 20 samples being independent, $X \sim \mathcal{B}(20, 0{,}032)$.

4. $P(X = 0) = 0{,}968^{20} \simeq 0.5218$. For further information,

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1) = 1 - 0{,}968^{20} - 20(0{,}032)0{,}968^{19} \simeq 0{,}1332.$$

Finally, $E(X) = 20 \times 0{,}032 = 0{,}64$. This expectation does not imply that a sample has a
fractional number of defects.

5. A common cause related to the lot could make defects dependent.
calculations are based on the independence given in the model; this hypothesis should be reconsidered
for another protocol.

E1. Total probabilities, $P(B) = 0{,}3 \times 0{,}8 + 0{,}7 \times 0{,}1 = 0{,}31$. Since $P(B) > 0$, $P_B(A) = \frac{P(A\cap B)}{P(B)} = \frac{0{,}24}{0{,}31} = \frac{24}{31}$.

E2. By the binomial formula, $P(X = 3) = \binom{8}{3}(\frac{1}{4})^3(\frac{3}{4})^5 = 56(\frac{1}{4})^3(\frac{3}{4})^5$. Parameters give $E(X) = \frac{8}{4} = 2$ and $V(X) = 8(\frac{1}{4})(\frac{3}{4}) = \frac{3}{2}$.

E3. The supplement to $X \geq 4$ is $X \leq 3$, so $P(X \geq 4) = 1 - F(3)$. For the interval, $X \leq 5$ values $X \leq 1$, therefore $P(2 \leq X \leq 5) = F(5) - F(1)$.

E4. No, in a usual game containing four aces. Without discount, get an ace in the first draw changes the probability of obtaining one in the second: the tests are not independent.
d·as can never be worth 5, while a binomial of parameters 5 and 0 $< p < 1$ gives a Positive probability at this value.

E5. The probability of at least one success is 1 $-(\frac{1}{2})^n$. The condition is equivalent to 2 $^n \geq 100$. Like two. $^6 = 64 < 100 \leq 128 = 2^7$, the first integer is 7. $\frac{127}{128} \simeq 0{,}99219$.

Corrected from DM13

1. X counts the success of twenty independent Bernoulli tests of parameter 0 ,8. Therefore $X \sim \mathcal{B}(20, 0{,}8)$, $E(X) = 20 \times 0{,}8 = 16$ and $V(X) = 20 \times 0{,}8 \times 0{,}2 = 3{,}2$.

2. The room lacks seats if $X > 18$, i.e. $X = 19$ or $X = 20$. These events being incompatible,
$$P(X > 18) = 20(0{,}8)^{19}(0{,}2) + (0{,}8)^{20} \simeq 0{,}069175.$$
The risk is about 6.92 %.

3. Number of refusals R equals 0 for $X \leq 18$, 1 for $X = 19$ and 2 for $X = 20$. By definition of expectation,
$$E(R) = P(X = 19) + 2P(X = 20) \simeq 0{,}0807045.$$
Revenues are 20 $\times 10 = 200$ and payments 30 R, therefore $G = 200 - 30R$. Thus $E(G) = 200 - 30E(R) \simeq 197{,}5789$ monetary units. The 30 already include all payments related to a refusal; do not withdraw yet 10.

4. With 18 reservations, no refusal is possible and the revenue is 18 $\times 10 = 180$, what be the number of present. The expectation of the gain with twenty reservations exceeds this amount d-about 17.58 monetary units.

5. The arrivals of people from the same group can be dependent.
and the consequences of a refusal are not summarized by the expectation of gain.
It is not enough to impose a decision alone.

Corrected work in the manual or an anal

The 100 choices are independent, with the same probability $p = 0.0714$. Therefore $X \sim \mathcal{B}(100; 0{,}0714)$.

$$P(X \leq 7) = \sum_{k=0}^{7} \binom{100}{k} (0{,}0714)^k (0{,}9286)^{100-k} \simeq 0{,}577.$$

Finally, $E(X) = 7{,}14$ and $V(X) = 100 \times 0{,}0714 \times 0{,}9286 = 6{,}630204 \simeq 6{,}63$.